

Week 1 Lecture Notes: Introduction to Quantum Computing

Cornell Quantum Computing Association - Fundamentals

1 Course Overview and Context (20 minutes)

1.1 Why Quantum Computing Matters

Quantum computing isn't just a faster version of classical computing – it's a fundamentally different way of processing information. For certain problems, quantum computers can provide exponential speedups over the best known classical algorithms. This means problems that would take classical computers longer than the age of the universe could potentially be solved by quantum computers in hours or days.

The applications span from breaking current cryptographic systems to simulating molecular interactions for drug discovery, from optimizing complex logistics to enhancing machine learning algorithms. Major tech companies and governments are investing billions because they recognize this isn't just an incremental improvement – it's a paradigm shift.

Current quantum computers are in the early stages – they're noisy, error-prone, and limited to specific problems. But we're just beginning to see “quantum advantage” for certain specialized tasks. The timeline for broader applications varies: quantum chemistry simulations might be practical in 5-10 years, optimization problems in 10-15 years, and breaking current cryptography in 15-25 years.

2 Building the Classical Foundation: The Coin Flip World (15 minutes)

2.1 Starting Simple: A Single Coin

Let's begin with something completely familiar – flipping a coin. In our classical world, when you flip a coin and it lands, it shows either heads or tails. Never both, never neither. This is a binary system with two definite outcomes.

We can represent this mathematically. Let's say heads = 0 and tails = 1. After the coin lands, we have certainty:

- State “heads” = 0
- State “tails” = 1

But let's be more precise about this mathematical representation. We can write these states as vectors:

The “heads” state:

$$\text{heads} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

The “tails” state:

$$\text{tails} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

These vectors capture something important: the coin is completely in one state (the “1” entry) and completely not in the other state (the “0” entry). This will connect directly to quantum notation when we introduce qubits.

Now, before the coin lands, we might not know the outcome. We could say “it’s 50% heads, 50% tails.” But this probability just represents our ignorance – the coin will land in a definite state, we just don’t know which one yet.

2.2 Multiple Coins: Exponential Growth

Now let’s consider flipping two coins. The possible outcomes are:

Coin 1: Heads, Coin 2: Heads $\rightarrow (0, 0)$ (1)

Coin 1: Heads, Coin 2: Tails $\rightarrow (0, 1)$ (2)

Coin 1: Tails, Coin 2: Heads $\rightarrow (1, 0)$ (3)

Coin 1: Tails, Coin 2: Tails $\rightarrow (1, 1)$ (4)

That’s 4 possible outcomes for 2 coins. With 3 coins, we get 8 outcomes. With n coins, we get 2^n possible outcomes. This exponential growth will become very important.

2.3 The Classical Computing Connection

This coin-flipping model is exactly how classical computers work. Instead of coins, we have bits. Instead of heads and tails, we have 0 and 1. A classical computer with n bits can store exactly one of 2^n possible values at any given time.

Your computer might have 8 gigabytes of RAM – that’s about 64 billion bits, which can store one of $2^{64 \text{ billion}}$ possible configurations. That’s an incomprehensibly large number, but it’s still just one definite configuration at a time.

2.4 Limitations of the Classical Approach

Here’s where we start to see limitations. Suppose you want to solve a problem where you need to check all possible arrangements of n coins. Maybe you’re trying to find a specific pattern, or the arrangement that satisfies some complicated condition.

Classically, you have to check each arrangement one by one:

- Check arrangement 1: $(0, 0, 0, \dots, 0)$
- Check arrangement 2: $(0, 0, 0, \dots, 1)$
- Check arrangement 3: $(0, 0, 1, \dots, 0)$
- ... and so on

With n coins, you need to check 2^n arrangements. For small n , this is fine. But as n grows, this becomes impossible. With just 300 coins, you’d have more arrangements to check than there are atoms in the observable universe.

3 Mathematical Foundations: From Bits to Qubits (8 minutes)

3.1 Quantum Notation and States

Now we're ready to make the transition to quantum systems. In quantum computing, we use special notation that connects directly to our classical vectors.

We write the quantum version of our classical states as:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (\text{the "heads" state}) \quad (5)$$

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (\text{the "tails" state}) \quad (6)$$

The notation $|0\rangle$ is called a “ket” and is read as “ket zero.” This is part of what’s called Dirac notation, and it’s the standard way to write quantum states.

These states $|0\rangle$ and $|1\rangle$ form what we call the *computational basis* or *standard basis*. Just like our classical coins, a classical bit must be in exactly one of these two states.

3.2 Introducing Superposition

But here’s where quantum systems become different. A quantum bit (qubit) can exist in a *superposition* of both $|0\rangle$ and $|1\rangle$ simultaneously:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

Here, α and β are complex numbers called probability amplitudes. This isn’t just mathematical notation – it represents a genuine physical state that has no classical analog.

In vector form, this becomes:

$$|\psi\rangle = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

3.3 Probabilities and Normalization

When we measure this quantum state, we get probabilistic outcomes:

- Probability of measuring $|0\rangle$: $|\alpha|^2$
- Probability of measuring $|1\rangle$: $|\beta|^2$

Since these probabilities must sum to 1 (we must get some outcome), we have the normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1$$

This constraint ensures that our quantum state represents valid probabilities.

3.4 Quantum Gates as Matrices

Just as classical computers use logic gates to manipulate bits, quantum computers use quantum gates to manipulate qubits. These gates can be represented as matrices that act on our quantum state vectors.

For example, a quantum NOT gate (called the X gate) flips $|0\rangle$ to $|1\rangle$ and vice versa:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

We can verify this works:

$$X|0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle \quad (7)$$

$$X|1\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle \quad (8)$$

This mathematical framework allows us to precisely describe and calculate quantum operations. In the coming weeks, we'll learn about many more quantum gates and how to combine them into quantum algorithms.

4 Introducing the Quantum Coin: A Different Kind of Object (10 minutes)

4.1 The Quantum Coin That Doesn't Land

Now imagine a different kind of coin – a quantum coin. When you flip this coin, instead of landing as heads or tails, it somehow keeps spinning forever. While it's spinning, it's not exactly heads and not exactly tails. It's somehow both.

This sounds impossible in our everyday world, but this is exactly how quantum objects behave. A quantum coin (which we'll call a “qubit”) can exist in what we call a “superposition” of heads and tails simultaneously.

This isn't just about our uncertainty. Remember, with a classical coin, our “50% heads, 50% tails” was just our ignorance – the coin had actually landed in a definite state. With a quantum coin, the coin genuinely hasn't “decided” yet. It's in a real physical state that's simultaneously both heads and tails.

Mathematically, we might write this balanced superposition as:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

The coefficients $\frac{1}{\sqrt{2}}$ are chosen so that $|\frac{1}{\sqrt{2}}|^2 + |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2} + \frac{1}{2} = 1$, satisfying our normalization condition.

4.2 Measuring the Quantum Coin

Here's where it gets even stranger. When you look at the quantum coin to see what state it's in, something dramatic happens. The act of looking – what we call “measurement” – forces the coin to “choose” heads or tails. The spinning stops, the superposition collapses, and you get a definite classical outcome.

If the quantum coin was in the balanced superposition above, then when you measure it, you get $|0\rangle$ 50% of the time and $|1\rangle$ 50% of the time. But before the measurement, it wasn't randomly $|0\rangle$ or $|1\rangle$ – it was genuinely both.

4.3 Creating Superposition

How do we create this magical spinning state? We need quantum gates that can put a classical state into superposition. One of the most important is what we'll call the Hadamard gate (we'll learn more about it next week):

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

When applied to $|0\rangle$:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

This creates our perfectly balanced quantum coin!

5 Multiple Quantum Coins: Exponential Possibilities (10 minutes)

5.1 Two Quantum Coins

Now let's consider two quantum coins, both in superposition. Something remarkable happens: the quantum system can exist in a superposition of all possible classical outcomes simultaneously.

While two classical coins give you one of four definite outcomes, two quantum coins in superposition can exist in a state that's simultaneously:

$$|\psi\rangle = \alpha_1|00\rangle + \alpha_2|01\rangle + \alpha_3|10\rangle + \alpha_4|11\rangle$$

This quantum system contains information about all four classical possibilities at once. The probabilities still sum to 1:

$$|\alpha_1|^2 + |\alpha_2|^2 + |\alpha_3|^2 + |\alpha_4|^2 = 1$$

5.2 Scaling Up: The Exponential Advantage

With n quantum coins in superposition, your quantum system can exist in a superposition of all 2^n classical outcomes simultaneously. This is fundamentally different from classical parallelism:

Classical approach: To work with 2^n possibilities, you need 2^n different classical computers, each checking one possibility.

Quantum approach: To work with 2^n possibilities, you need just n quantum coins in superposition.

The resources scale logarithmically instead of exponentially. To work with a million possibilities, you need about 20 quantum coins instead of a million classical computers.

5.3 The Catch: Measurement Collapse

But here's the catch that prevents quantum computers from being magical: when you measure your n quantum coins to get an answer, all those exponentially many possibilities collapse down to just one classical outcome.

You can't directly read out all 2^n pieces of information. You get just one answer, chosen probabilistically based on the quantum amplitudes.

5.4 Why Only Certain Problems Benefit

Not every problem can exploit this quantum advantage. You need problems where:

1. You're searching through an exponentially large space
2. You can design interference patterns that amplify correct answers
3. The structure of the problem allows quantum interference to work

This is why quantum computers excel at specific problems like factoring numbers, searching databases, and simulating quantum systems, but don't help much with tasks like word processing or web browsing. In the coming weeks, we'll see exactly how algorithms like Shor's factoring algorithm and Grover's search algorithm exploit these principles.

6 Building Toward Quantum Computation (7 minutes)

6.1 What We Need to Make This Work

To build quantum algorithms, we need:

1. **Ways to create superposition:** Starting with classical states and putting them into quantum superposition
2. **Ways to manipulate superposition:** Operations that change quantum states in useful ways
3. **Ways to create interference:** Making wrong answers cancel out and right answers amplify
4. **Ways to measure:** Converting quantum results back to classical information

Over the next few weeks, we'll learn how to do each of these things systematically.

6.2 Mathematical Tools We'll Develop

Next week, we'll complete our mathematical foundation by learning about:

- The Bloch sphere: A geometric way to visualize qubit states
- More quantum gates: The building blocks of quantum circuits
- The Born rule: The precise mathematical description of quantum measurement
- Multiple qubit systems: How to work with many qubits simultaneously